A Comparative Analysis of Capacitor-Less LDO Topologies in 130nm CMOS Technology

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***Abstract*—This paper presents the design and comparison of three capacitor-less (CL) PMOS low dropout (LDO) regulators implemented in a 130nm CMOS technology. Each LDO topology was designed using a different OTA (5T OTA, Current Mirror OTA, Telescopic OTA). All three LDO configurations drive a 50 pF load, deliver a load current of $500\,\mu\mathrm{A}$, support programmable output voltages ($2.3\,\mathrm{V}$, $2.1\,\mathrm{V}$ and $1.6\,\mathrm{V}$), and have a consumption strictly below $100\,\mu\mathrm{A}$. Simulation results demonstrate that the 5T OTA provides the most balanced final results, whereas the Telescopic OTA maximizes loop gain at the sever expense of the Error Amplifier's output voltage swing. Furthermore, the Current Mirror OTA is shown to be incapable of satisfying the target specifications while keeping all the transistors in saturation.**

I. Introduction and Theoretical Framework

Low dropout (LDO) voltage regulators are one of the most essential building blocks in power-management systems. Specially with today's increasing demand for system-on-chip solutions, many mixed-signal systems utilize LDO regulators to deliver local supply voltages with reduced noise.

This work will focus on designing three individual LDO architectures. Each LDO topology must meet the proposed specifications that are presented in Table I. The general LDO schematic is also shown in Figure 1.

TABLE I
Technical LDO Target Specifications

Parameter	Proposed Value
Power Supply Voltage	$3\,\mathrm{V}$
Bias Current I_{bias}	$5\,\mu\mathrm{A}$
Load Current I_L	$500\,\mu\mathrm{A}$
Load Capacitance C_L	$50\,\mathrm{pF}$
Reference Voltage V_{REF}	$1.2\,\mathrm{V} \pm 1\,\%$
Output Voltage V_{OUT}	$2.3\,\mathrm{V}$
Trim Voltages	$2.1\,\mathrm{V}$, $1.6\,\mathrm{V}$
Compensation Capacitor C_c	$<7\,\mathrm{pF}$
Phase Margin @ No-Load	$>60°$
Phase Margin @ Full-Load	$>70°$
PSRR @ 1 Hz	$>65\,\mathrm{dB}$
PSRR @ 1 MHz	$>2\,\mathrm{dB}$
PSRR @ 10 MHz	$>2\,\mathrm{dB}$
Output Referred Noise @ 10 Hz	$<-115\,\mathrm{dBV}/\sqrt{\mathrm{Hz}}$
Output Referred Noise @ 1 kHz	$<-125\,\mathrm{dBV}/\sqrt{\mathrm{Hz}}$
Output Referred Noise @ 10 kHz	$<-134\,\mathrm{dBV}/\sqrt{\mathrm{Hz}}$
Output Referred Noise @ 100 kHz	$<-143\,\mathrm{dBV}/\sqrt{\mathrm{Hz}}$
Output Referred Noise @ 1 MHz	$<-155\,\mathrm{dBV}/\sqrt{\mathrm{Hz}}$

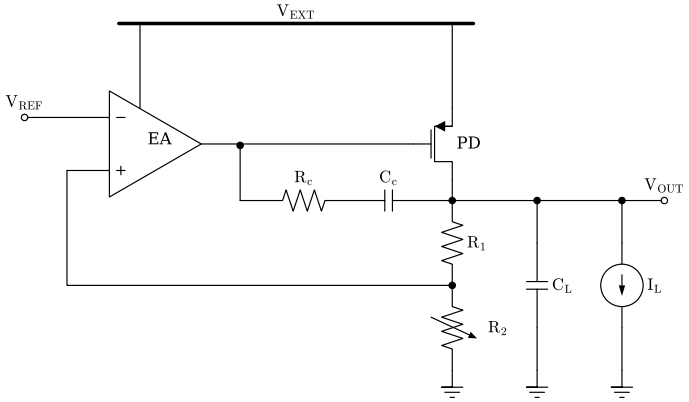


Fig. 1. General LDO Schematic

As shown in Figure 1, a PMOS LDO is made of an Error Amplifier (EA), a PMOS Pass Device (PD), two feedback resistors R_1 and R_2 for setting the output voltage and a compensation network consisting of a resistor R_c and a capacitor C_c. In this work only one-stage EAs will be presented and analyzed.

A. Output Voltage

Assuming that the EA has infinite DC gain (therefore $A_{EA,0} \to \infty$), LDO's gain is given by:

$$\frac{V_{OUT}}{V_{REF}} = \frac{A_{EA,0}}{1 + A_{EA,0}\beta} \xrightarrow{A_{EA,0}\to\infty} \frac{V_{OUT}}{V_{REF}} = \frac{1}{\beta} \tag{1}$$

where β is the feedback factor set by $1 + \frac{R_1}{R_2}$. Therefore:

$$\frac{V_{OUT}}{V_{REF}} = \frac{R_1}{R_1 + R_2} \to \frac{R_1}{R_2} = \frac{V_{REF}}{V_{OUT}} - 1 \tag{2}$$

B. Stability

A small signal model of the LDO schematic can be seen in Figure 2 where the EA's transconductance, output resistance and output load are represented by $g_{m,EA}$, R_1 and C_1 respectively. Similarly, $g_{m,p}$, R_2 and C_2 represent the same values for the PD. The LDO's transfer function has two poles (given that we have a one stage EA), ω_{p1} (dominant) and ω_{p2} (non-dominant) and a zero ω_z, all of which can be approximated with:

$$\omega_{p1} = \frac{1}{g_{m,p} R_1 R_2 C_c} \tag{3}$$

$$\omega_{p2} = \frac{g_{m,p}}{C_1 + C_2} \tag{4}$$

$$\omega_z = \frac{-1}{C_c(1/g_{m,p} - R_c)} \tag{5}$$

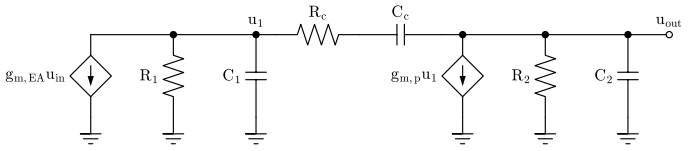


Fig. 2. Small Signal Model of LDO

Even though unity gain bandwidth is not a required specification, one could find the unity gain frequency of the LDO using:

$$UGF \cong \frac{\beta g_{m,EA}}{C_2} \tag{6}$$

C. Power Supply Rejection Ratio (PSRR)

Power supply rejection refers to the amount of voltage ripple at the output of the LDO coming from the input voltage and therefore is of great importance. The PSR transfer function of a LDO regulator strongly depends on the type of the EA and whether the input transistors of the EA are NMOS or PMOS (Type-A or Type-B EA respectively). For a PMOS PD, Type-A EAs give the best PSRR results at low frequencies [2]. Therefore, all the LDOs presented in this work utilize Type-A EAs and their PSRR can be approximated by:

$$|PSRR_{DC}| \cong \frac{1}{\beta A_{EA,0} g_{m,p} r_{ds,p}} \tag{7}$$

At higher frequencies, the compensation resistor along with the sizing of the PD can help in achieving the target PSRR requirements.

D. Noise

Noise in LDO regulators refer to the flicker and thermal noise generated by transistors and resistors. Aside from that, the reference voltage can also contribute to the total noise in LDO. If $S_{n,ref}(f)$, $S_{n,EA}(f)$, $S_{n,PD}(f)$, $S_{n,R1}(f)$ and $S_{n,R2}(f)$ represent the noise power spectral denisty of the voltage reference, error amplifier, pass device and feedback resistors respectively, then the total output noise power spectral density of the LDO regulator is [2]:

$$\begin{aligned} S_{n,o}(f) \cong (S_{n,ref}(f) + S_{n,EA}(f)) \left(1 + \frac{R_1}{R_2}\right)^2 \\ + S_{n,R2}(f)(\frac{R_1}{R_2})^2 + S_{n,R_1}(f) \end{aligned} \tag{8}$$

The EA and feedback resistors noise are typically the dominant sources of a LDO regulator noise.

II. 5T OTA-Based LDO

The 5-transistor OTA used as EA in this LDO variation can be seen in Figure 3. 5T OTA is a simple differential pair with a gain of:

$$A_{5T} = \frac{g_{m,12}}{g_{ds,2} + g_{ds,4}} \tag{9}$$

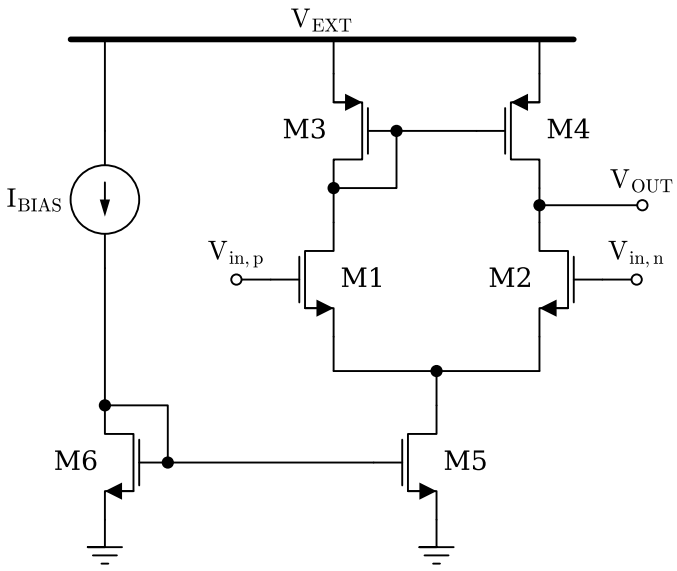


Fig. 3. 5T OTA

whereas the output impedance:

$$g_{5T,out} = g_{ds,2} + g_{ds,4} \tag{10}$$

For achieving the PSRR specifications, we must maximize the OTA's loop gain. This can be achieved by increasing the transconductance of the NMOS input pair and the output resistance. Also, large transistors will reduce the Flicker noise leading to reduced output referred noise at lower frequencies. Table II shows the sizing of transistors for this LDO variation.

TABLE II
Device Sizing for the 5T OTA LDO

Device	Value / Sizing	Unit	
Error Amplifier	M_1, M_2 (Input Pair)	42.0/5.0	µm
	M_3, M_4 (Active Load)	3.8/5.0	µm
	M_5 (Tail Current)	12.0/5.0	µm
Power Stage	M_{PD} (Pass Device)	20.0/0.14	µm
Passive Network	R_1 (Divider)	15.00	kΩ
	R_{21} (Divider)	16.36	kΩ
	R_{22} (Divider)	20.00	kΩ
	R_{23} (Divider)	45.00	kΩ
	R_c (Compensation)	15.00	kΩ
	C_c (Compensation)	7.00	pF

Simulations prove that the width of the PMOS active load can directly affect the PSRR of LDO and as a result, a small PMOS active load was chosen. For reducing the noise at higher frequencies (Thermal noise), we increased the tail current of the differential pair, leading to the final value presented in the previous table.

III. Telescopic OTA-Based LDO

IV. Current Mirror OTA-Based LDO

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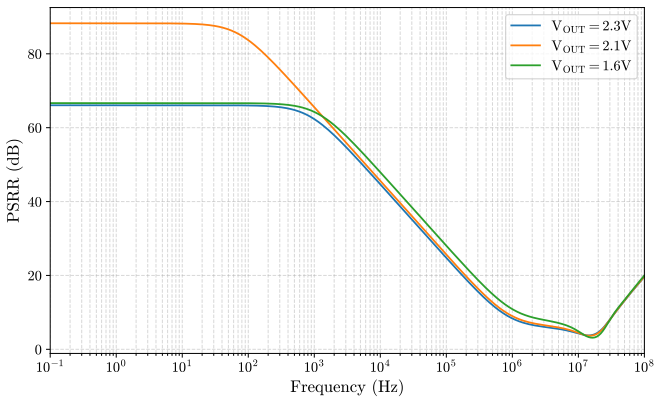


Fig. 4. 5T OTA-Based LDO's PSRR

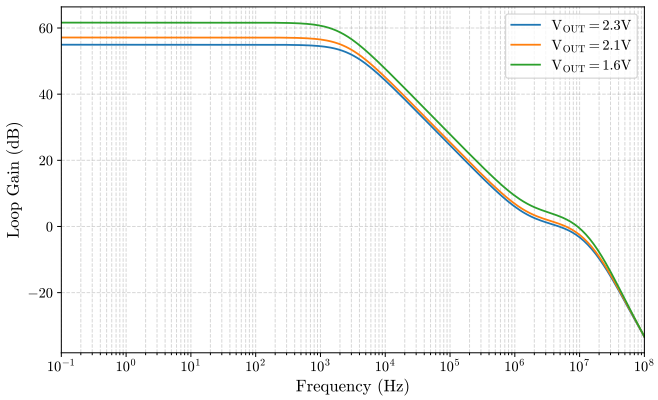


Fig. 5. 5T OTA-Based LDO's Loop Gain

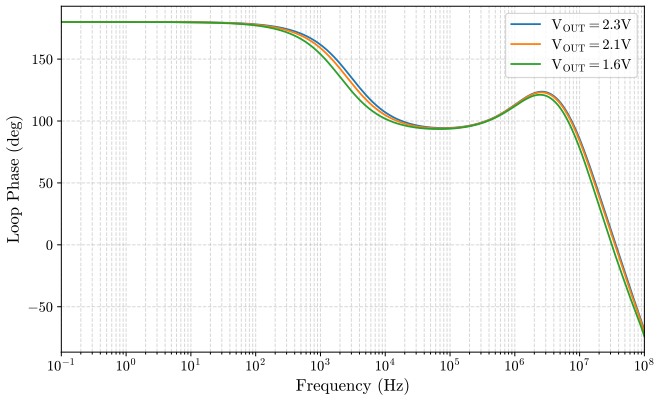


Fig. 6. 5T OTA-Based LDO's Loop Phase

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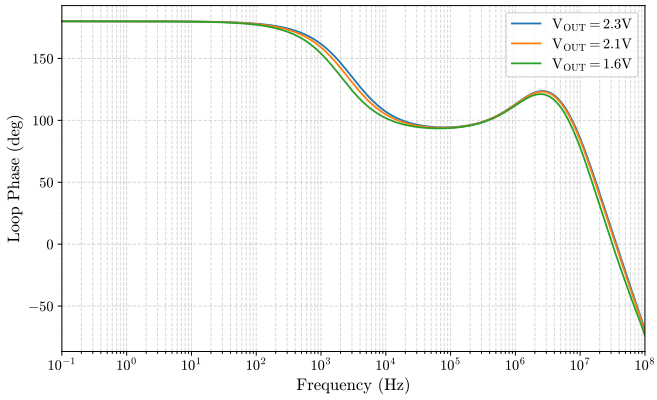


Fig. 7. 5T OTA-Based LDO's Loop Phase

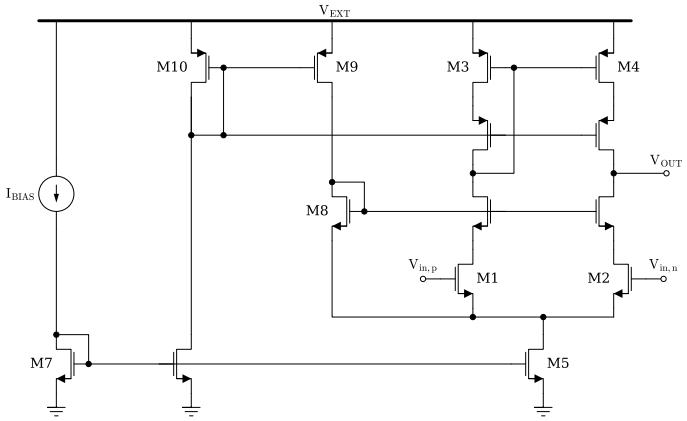


Fig. 8. Telescopic OTA

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